Compile time

Generated once per rule

User rule (matrix example)

```
(dot[@d]
  (transpose[@tr] A) B)
=> (dot[auto] A B)
```

auto: generate attributes

Operation semantics

```
transpose
dot
```

TEPL compiler

Synthesize + verify

Generated attribute function

`f(p, q) → replacement dot attrs`

`p = @tr.permutation; q = @d`

Runtime

Evaluate on each match

Captured attrs

`p = [1, 0]`

q: left axis = 0

→ **Apply f**

`f(p, q)`

→ **New dot attrs**

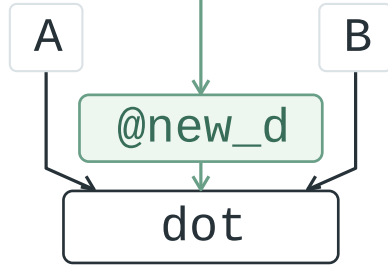
left axis = 1

@new_d

Apply replacement

`(dot[@new_d] A B)`

Transpose removed



Only the left contraction axis is shown.